COST SENSITIVE LEARNING FOR MYOCARDIAL INFARCTION MORTALITY PREDICTION CRISP-DM PROCESS REPORTs

BUSINESS UNDERSTANDING

Problems of real-life complexity are needed to test and compare various data mining and pattern recognition methods. The proposed database can be used to solve two practically important problems: predicting complications of Myocardial Infarction (MI) based on information about the patient (i) at the time of admission and (ii) on the third day of the hospital period. Another important group of tasks is phenotyping of disease (cluster analysis), dynamic phenotyping (filament extraction and identification of disease trajectories) and visualisation (disease mapping).

MI is one of the most challenging problems of modern medicine. Acute myocardial infarction is associated with high mortality in the first year after it. The incidence of MI remains high in all countries. This is especially true for the urban population of highly developed countries, which is exposed to chronic stress factors, irregular and not always balanced nutrition. In the United States, for example, more than a million people suffer from MI every year, and 200-300 thousand of them die from acute MI before arriving at the hospital.

The course of the disease in patients with MI is different. MI can occur without complications or with complications that do not worsen the long-term prognosis. At the same time, about half of patients in the acute and subacute periods have complications that lead to worsening of the disease and even death. Even an experienced specialist can not always foresee the development of these complications. In this regard, predicting complications of myocardial infarction in order to timely carry out the necessary preventive measures is an important task.

DATA UNDERSTANDING: *Data Description Report*

- **Location:** The original dataset is available at https://archive-beta.ics.uci.edu/dataset/579/myocardial+infarction+complications
- **Method used to acquire:** Download the dataset using the ucimlrepo library, which provides a simple way to interact with uci machine learning resources.
- **Format:** csv file
- **Quantity:** Rows (1700) - Columns (124)
- **Identities of the fields:** The dataset analyzed here was collected in the Krasnoyarsk Interdistrict Clinical Hospital (Russia) in 1992-1995 years, but has only recently been deposited to the public domain. It contains information about 1700 patients and 110 features characterizing the clinical phenotypes and 12 features representing possible complications of the myocardial infarction disease. In general columns 2-112 can be used as input data for prediction. Possible complications (outputs) are listed in columns 113-124. There are four possible time moments for complication prediction on base of the information known at:
    - the time of admission to hospital - all input columns (2-112) except 93, 94, 95, 100, 101, 102, 103, 104, 105 can be used for prediction;
    - the end of the first day (24 hours after admission to the hospital): all input columns (2-112) except 94, 95, 101, 102, 104, 105 can be used for prediction;
    - the end of the second day (48 hours after admission to the hospital) all input columns (2-112) except 95, 102, 105 can be used for prediction;
    - the end of the third day (72 hours after admission to the hospital) all input columns (2-112) can be used for prediction.
- **Surfaces features:** ID, AGE, SEX, INF_ANAM, STENOK_AN, FK_STENOK, IBS_POST, IBS_NASL, GB, SIM_GIPERT, DLIT_AG, ZSN_A, nr_11, nr_01, nr_02, nr_03, nr_04, nr_07, nr_08, np_01, np_04, np_05, np_07, np_08, np_09, np_10, endocr_01, endocr_02, endocr_03, zab_leg_01, zab_leg_02, zab_leg_03, zab_leg_04, zab_leg_06, S_AD_KBRIG, D_AD_KBRIG, S_AD_ORIT, D_AD_ORIT, O_L_POST, K_SH_POST, MP_TP_POST, SVT_POST, GT_POST, FIB_G_POST, ant_im, lat_im, inf_im, post_im, IM_PG_P, ritm_ecg_p_01, ritm_ecg_p_02, ritm_ecg_p_04, ritm_ecg_p_06, ritm_ecg_p_07, ritm_ecg_p_08, n_r_ecg_p_01, n_r_ecg_p_02, n_r_ecg_p_03, n_r_ecg_p_04, n_r_ecg_p_05, n_r_ecg_p_06, n_r_ecg_p_08, n_r_ecg_p_09, n_r_ecg_p_10, n_p_ecg_p_01, n_p_ecg_p_03, n_p_ecg_p_04, n_p_ecg_p_05, n_p_ecg_p_06, n_p_ecg_p_07, n_p_ecg_p_08, n_p_ecg_p_09, n_p_ecg_p_10, n_p_ecg_p_11, n_p_ecg_p_12, fibr_ter_01, fibr_ter_02, fibr_ter_03, fibr_ter_05, fibr_ter_06, fibr_ter_07, fibr_ter_08, GIPO_K, K_BLOOD, GIPER_NA, NA_BLOOD, ALT_BLOOD, AST_BLOOD, KFK_BLOOD, L_BLOOD, ROE, TIME_B_S, R_AB_1_n, R_AB_2_n, R_AB_3_n, NA_KB, NOT_NA_KB, LID_KB, NITR_S, NA_R_1_n, NA_R_2_n, NA_R_3_n, NOT_NA_1_n, NOT_NA_2_n, NOT_NA_3_n, LID_S_n, B_BLOK_S_n, ANT_CA_S_n, GEPAR_S_n, ASP_S_n, TIKL_S_n, TRENT_S_n,

    - ***Quantitative Variables***
        - **Discrete:** AGE, S_AD_KBRIG, D_AD_KBRIG, S_AD_ORIT, D_AD_ORIT
        - **Continuous:** K_BLOOD, NA_BLOOD, ALT_BLOOD, AST_BLOOD, KFK_BLOOD, L_BLOOD, ROE
    - ***Categorical Variables***
        - **Nominal**
            - ***Binary*:** SEX, IBS_NASL, SIM_GIPERT, nr_11, nr_01, nr_02, nr_03, nr_04, nr_07, nr_08, np_01, np_04, np_05, np_07, np_08, np_09, np_10, endocr_01, endocr_02, endocr_03, zab_leg_01, zab_leg_02, zab_leg_03, zab_leg_04, zab_leg_06, O_L_POST, K_SH_POST, MP_TP_POST, SVT_POST, GT_POST, FIB_G_POST, IM_PG_P, ritm_ecg_p_01, ritm_ecg_p_02, ritm_ecg_p_04, ritm_ecg_p_06, ritm_ecg_p_07, ritm_ecg_p_08, n_r_ecg_p_01, n_r_ecg_p_02, n_r_ecg_p_03, n_r_ecg_p_04, n_r_ecg_p_05, n_r_ecg_p_06, n_r_ecg_p_08, n_r_ecg_p_09, n_r_ecg_p_10, n_p_ecg_p_01, n_p_ecg_p_03, n_p_ecg_p_04, n_p_ecg_p_05, n_p_ecg_p_06, n_p_ecg_p_07, n_p_ecg_p_08, n_p_ecg_p_09, n_p_ecg_p_10, n_p_ecg_p_11, n_p_ecg_p_12, fibr_ter_01, fibr_ter_02, fibr_ter_03, fibr_ter_05, fibr_ter_06, fibr_ter_07, fibr_ter_08, GIPO_K, GIPER_NA, NA_KB, NOT_NA_KB, LID_KB, NITR_S, LID_S_n, B_BLOK_S_n, ANT_CA_S_n, GEPAR_S_n, ASP_S_n,

TIKL_S_n, TRENT_S_n, FIBR_PREDS, PREDS_TAH, JELUD_TAH, FIBR_JELUD, A_V_BLOK, OTEK_LANC, RAZRIV, DRESSLER, ZSN, REC_IM, P_IM_STEN
    - ***General***: LET_IS, ID, IBS_POST
  - **Ordinal**: INF_ANAM, STENOK_AN, FK_STENOK, GB, DLIT_AG, ZSN_A, TIME_B_S, R_AB_1_n, R_AB_2_n, R_AB_3_n, NA_R_1_n, NA_R_2_n, NA_R_3_n, NOT_NA_1_n, NOT_NA_2_n, NOT_NA_3_n, ant_im, lat_im, inf_im, post_im

- **Targets:** FIBR_PREDS, PREDS_TAH, JELUD_TAH, FIBR_JELUD, A_V_BLOK, OTEK_LANC, RAZRIV, DRESSLER, ZSN, REC_IM, P_IM_STEN, LET_IS

  - ***Quantitative Variables***
    - **Discrete: -**
    - **Continuous: -**
  - ***Categorical Variables***
    - **Nominal**
      - ***Binary***: FIBR_PREDS, PREDS_TAH, JELUD_TAH, FIBR_JELUD, A_V_BLOK, OTEK_LANC, RAZRIV, DRESSLER, ZSN, REC_IM, P_IM_STEN
      - ***General***: LET_IS
    - **Ordinal: -**

- **Features and Target by time moments for complication prediction:**
  - **admission**
    - **included:** ID, AGE, SEX, INF_ANAM, STENOK_AN, FK_STENOK, IBS_POST, IBS_NASL, GB, SIM_GIPERT, DLIT_AG, ZSN_A, nr_11, nr_01, nr_02, nr_03, nr_04, nr_07, nr_08, np_01, np_04, np_05, np_07, np_08, np_09, np_10, endocr_01, endocr_02, endocr_03, zab_leg_01, zab_leg_02, zab_leg_03, zab_leg_04, zab_leg_06, S_AD_KBRIG, D_AD_KBRIG, S_AD_ORIT, D_AD_ORIT, O_L_POST, K_SH_POST, MP_TP_POST, SVT_POST, GT_POST, FIB_G_POST, ant_im, lat_im, inf_im, post_im, IM_PG_P, ritm_ecg_p_01, ritm_ecg_p_02, ritm_ecg_p_04, ritm_ecg_p_06, ritm_ecg_p_07, ritm_ecg_p_08, n_r_ecg_p_01, n_r_ecg_p_02, n_r_ecg_p_03, n_r_ecg_p_04, n_r_ecg_p_05, n_r_ecg_p_06, n_r_ecg_p_08, n_r_ecg_p_09, n_r_ecg_p_10, n_p_ecg_p_01, n_p_ecg_p_03, n_p_ecg_p_04, n_p_ecg_p_05, n_p_ecg_p_06, n_p_ecg_p_07, n_p_ecg_p_08, n_p_ecg_p_09, n_p_ecg_p_10, n_p_ecg_p_11, n_p_ecg_p_12, fibr_ter_01, fibr_ter_02, fibr_ter_03, fibr_ter_05, fibr_ter_06, fibr_ter_07, fibr_ter_08, GIPO_K, K_BLOOD, GIPER_NA, NA_BLOOD, ALT_BLOOD, AST_BLOOD, KFK_BLOOD, L_BLOOD, ROE, TIME_B_S, NA_KB, NOT_NA_KB, LID_KB, NITR_S, LID_S_n, B_BLOK_S_n, ANT_CA_S_n, GEPAR_S_n, ASP_S_n, TIKL_S_n, TRENT_S_n, FIBR_PREDS, PREDS_TAH, JELUD_TAH, FIBR_JELUD, A_V_BLOK, OTEK_LANC, RAZRIV, DRESSLER, ZSN, REC_IM, P_IM_STEN, LET_IS
    - **excluded:** R_AB_1_n, R_AB_2_n, R_AB_3_n, NA_R_1_n, NA_R_2_n, NA_R_3_n, NOT_NA_1_n, NOT_NA_2_n, NOT_NA_3_n

  - **end of first day (after 24 hours)**
    - **included:** ID, AGE, SEX, INF_ANAM, STENOK_AN, FK_STENOK, IBS_POST, IBS_NASL, GB, SIM_GIPERT, DLIT_AG, ZSN_A, nr_11, nr_01, nr_02, nr_03, nr_04, nr_07, nr_08, np_01, np_04, np_05, np_07, np_08, np_09, np_10, endocr_01, endocr_02, endocr_03, zab_leg_01, zab_leg_02, zab_leg_03, zab_leg_04, zab_leg_06, S_AD_KBRIG, D_AD_KBRIG, S_AD_ORIT, D_AD_ORIT, O_L_POST, K_SH_POST, MP_TP_POST, SVT_POST, GT_POST, FIB_G_POST, ant_im, lat_im, inf_im, post_im, IM_PG_P, ritm_ecg_p_01, ritm_ecg_p_02, ritm_ecg_p_04, ritm_ecg_p_06, ritm_ecg_p_07, ritm_ecg_p_08, n_r_ecg_p_01, n_r_ecg_p_02, n_r_ecg_p_03, n_r_ecg_p_04, n_r_ecg_p_05, n_r_ecg_p_06, n_r_ecg_p_08, n_r_ecg_p_09, n_r_ecg_p_10, n_p_ecg_p_01, n_p_ecg_p_03, n_p_ecg_p_04, n_p_ecg_p_05, n_p_ecg_p_06, n_p_ecg_p_07, n_p_ecg_p_08, n_p_ecg_p_09, n_p_ecg_p_10, n_p_ecg_p_11, n_p_ecg_p_12, fibr_ter_01, fibr_ter_02, fibr_ter_03, fibr_ter_05, fibr_ter_06, fibr_ter_07, fibr_ter_08, GIPO_K, K_BLOOD, GIPER_NA, NA_BLOOD, ALT_BLOOD, AST_BLOOD, KFK_BLOOD, L_BLOOD, ROE, TIME_B_S, R_AB_1_n, NA_KB, NOT_NA_KB, LID_KB, NITR_S,

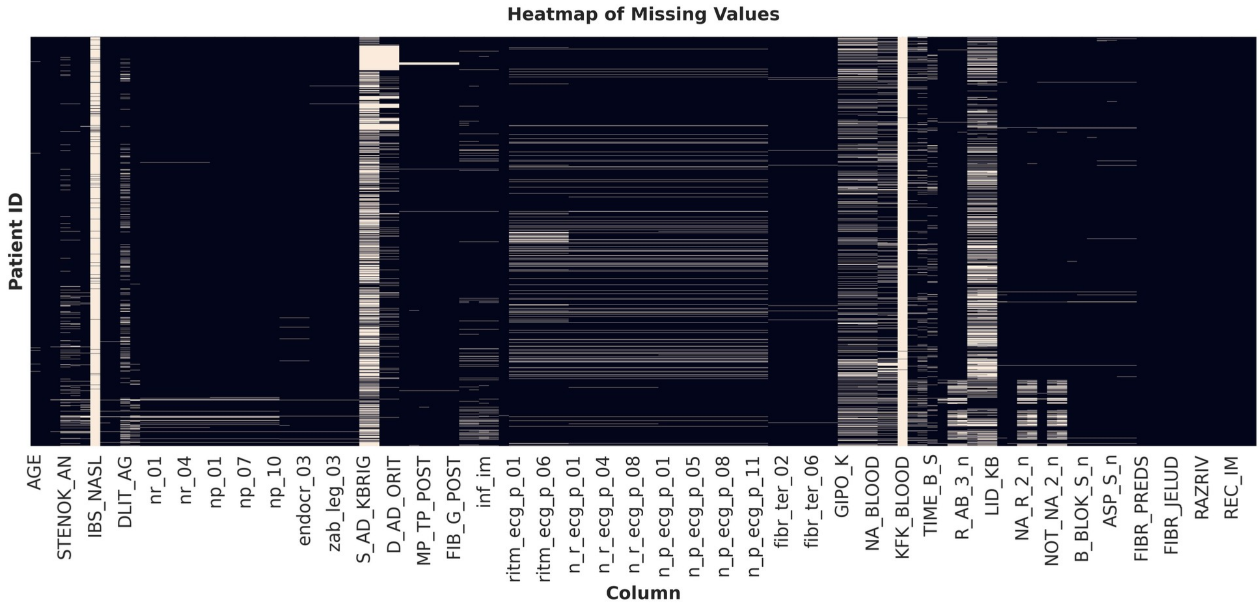
NA_R_1_n, NOT_NA_1_n, LID_S_n, B_BLOK_S_n, ANT_CA_S_n, GEPAR_S_n, ASP_S_n, TIKL_S_n, TRENT_S_n, FIBR_PREDS, PREDS_TAH, JELUD_TAH, FIBR_JELUD, A_V_BLOK, OTEK_LANC, RAZRIV, DRESSLER, ZSN, REC_IM, P_IM_STEN, LET_IS
    - **excluded:** R_AB_2_n, R_AB_3_n, NA_R_2_n, NA_R_3_n, NOT_NA_2_n, NOT_NA_3_n

- **end of second day (after 48 hours)**
    - **included:** ID, AGE, SEX, INF_ANAM, STENOK_AN, FK_STENOK, IBS_POST, IBS_NASL, GB, SIM_GIPERT, DLIT_AG, ZSN_A, nr_11, nr_01, nr_02, nr_03, nr_04, nr_07, nr_08, np_01, np_04, np_05, np_07, np_08, np_09, np_10, endocr_01, endocr_02, endocr_03, zab_leg_01, zab_leg_02, zab_leg_03, zab_leg_04, zab_leg_06, S_AD_KBRIG, D_AD_KBRIG, S_AD_ORIT, D_AD_ORIT, O_L_POST, K_SH_POST, MP_TP_POST, SVT_POST, GT_POST, FIB_G_POST, ant_im, lat_im, inf_im, post_im, IM_PG_P, ritm_ecg_p_01, ritm_ecg_p_02, ritm_ecg_p_04, ritm_ecg_p_06, ritm_ecg_p_07, ritm_ecg_p_08, n_r_ecg_p_01, n_r_ecg_p_02, n_r_ecg_p_03, n_r_ecg_p_04, n_r_ecg_p_05, n_r_ecg_p_06, n_r_ecg_p_08, n_r_ecg_p_09, n_r_ecg_p_10, n_p_ecg_p_01, n_p_ecg_p_03, n_p_ecg_p_04, n_p_ecg_p_05, n_p_ecg_p_06, n_p_ecg_p_07, n_p_ecg_p_08, n_p_ecg_p_09, n_p_ecg_p_10, n_p_ecg_p_11, n_p_ecg_p_12, fibr_ter_01, fibr_ter_02, fibr_ter_03, fibr_ter_05, fibr_ter_06, fibr_ter_07, fibr_ter_08, GIPO_K, K_BLOOD, GIPER_NA, NA_BLOOD, ALT_BLOOD, AST_BLOOD, KFK_BLOOD, L_BLOOD, ROE, TIME_B_S, R_AB_1_n, R_AB_2_n, NA_KB, NOT_NA_KB, LID_KB, NITR_S, NA_R_1_n, NA_R_2_n, NOT_NA_1_n, NOT_NA_2_n, LID_S_n, B_BLOK_S_n, ANT_CA_S_n, GEPAR_S_n, ASP_S_n, TIKL_S_n, TRENT_S_n, FIBR_PREDS, PREDS_TAH, JELUD_TAH, FIBR_JELUD, A_V_BLOK, OTEK_LANC, RAZRIV, DRESSLER, ZSN, REC_IM, P_IM_STEN, LET_IS
    - **excluded:** R_AB_3_n, NA_R_3_n, NOT_NA_3_n

- **end of third day (after 72 hours)**
    - **included:** ALL
    - **excluded: -**

DATA UNDERSTANDING: *Data Quality Report*

- **Missing values Analysis**



  - ***Row-Level Analysis:*** Total rows=1700, rows with ≥1 missing value=1700/1700 (100.00/100.00%). First of all, we quantify the general missngness in the dataset: the best-case record completeness has 0.81% of missing values; the worst-case record has 48.78% of missing values; the average quantity of missing values is 7.64%. Since 100% of rows have at least one missing value, the next step is to measure severity of missingness per row. We fix different threshold of missingness to assess how many rows have a level of missingness greater than the threshold: with a threshold > 5% we have 992/1700 record with a level of missingness greater than 5%; with a threshold > 20% we have 134/1700 record with a level of missingness greater than 5%; with a threshold > 50% we have 0/1700 record with a level of missingness greater than 50%.

  - ***Column-Level Analysis*:** We check the percentage of missingness in each columns to identify columns with an high level of missing values (to delete) and columns with moderate/low level of missing values (imputation). We create different bins to identify how many columns have a level of missing values which falls in those bins:
    - (0-1] % → 37 columns
    - (1-5] % → 21 columns
    - (5-10] % → 35 columns
    - (10-20] % → 6 columns
    - (20-50] % → 7 columns
    - (50-80] % → 2 columns
    - (80-100] % → 2 columns

In the data cleaning phase we will address the problem of missing values to select what record will be deleted, what features will be deleted, what missing values could be handle by imputation, what type of imputation is the most proper one.

- **Duplicated Row:** Number of duplicate rows: 0/1700 (0.000%). Nothing to do.

- **Class Balance:** In general, the dataset presents different possible targets, each of these representing one possible complication, with an additional target representing the lethal outcome. We report the distribution of patients for each possible target and value:

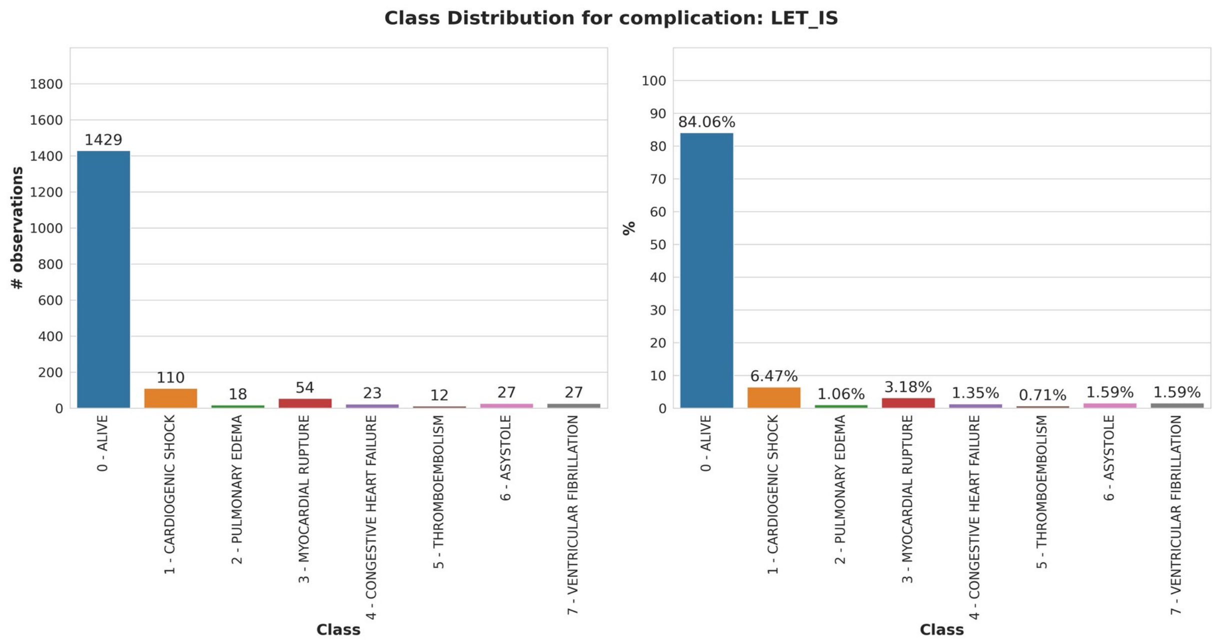
```
================================================================================
SAMPLES DISTRIBUTION BY: FIBR_PREDS
================================================================================
      -Class 0 - NO: 1530 (90.000%)
      -Class 1 - YES: 170 (10.000%)
================================================================================
SAMPLES DISTRIBUTION BY: PREDS_TAH
================================================================================
      -Class 0 - NO: 1680 (98.824%)
      -Class 1 - YES:  20 (1.176%)
================================================================================
SAMPLES DISTRIBUTION BY: JELUD_TAH
================================================================================
      -Class 0 - NO: 1658 (97.529%)
      -Class 1 - YES:  42 (2.471%)
================================================================================
SAMPLES DISTRIBUTION BY: A_V_BLOK
================================================================================
      -Class 0 - NO: 1643 (96.647%)
      -Class 1 - YES:  57 (3.353%)
================================================================================
SAMPLES DISTRIBUTION BY: OTEK_LANC
================================================================================
      -Class 0 - NO: 1541 (90.647%)
      -Class 1 - YES: 159 (9.353%)
================================================================================
SAMPLES DISTRIBUTION BY: RAZRIV
================================================================================
      -Class 0 - NO: 1646 (96.824%)
      -Class 1 - YES:  54 (3.176%)
================================================================================
SAMPLES DISTRIBUTION BY: DRESSLER
================================================================================
      -Class 0 - NO: 1625 (95.588%)
      -Class 1 - YES:  75 (4.412%)
================================================================================
SAMPLES DISTRIBUTION BY: ZSN
================================================================================
      -Class 0 - NO: 1306 (76.824%)
      -Class 1 - YES: 394 (23.176%)
================================================================================
SAMPLES DISTRIBUTION BY: REC_IM
================================================================================
      -Class 0 - NO: 1541 (90.647%)
      -Class 1 - YES: 159  (9.353%)
================================================================================
SAMPLES DISTRIBUTION BY: P_IM_STEN
================================================================================
      -Class 0 - NO: 1552 (91.294%)
      -Class 1 - YES: 148  (8.706%)
```

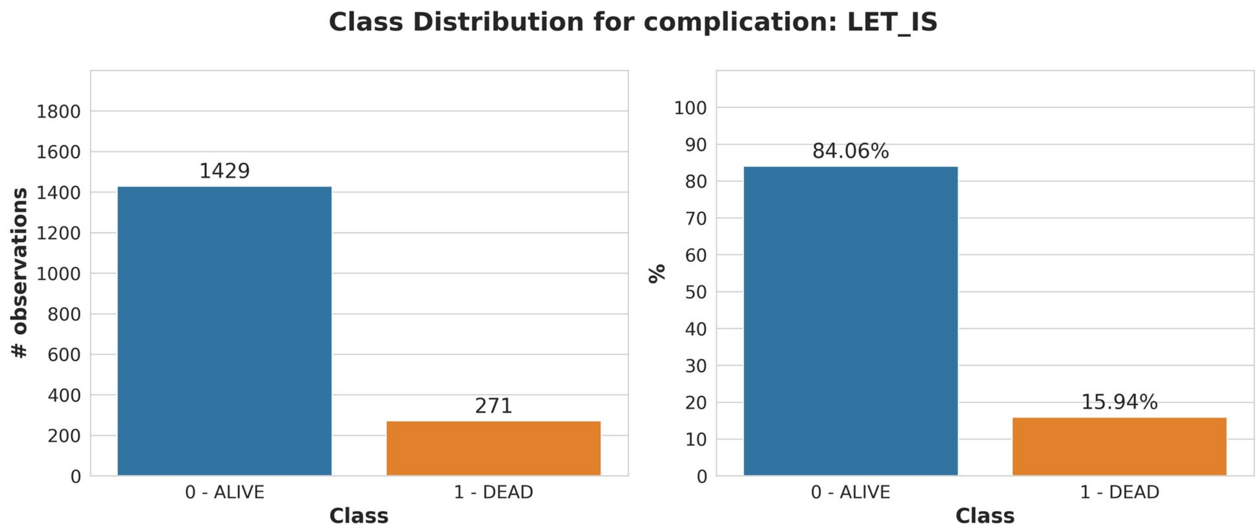
```
================================================================================
SAMPLES DISTRIBUTION BY: LET_IS
================================================================================
      -Class 0 - ALIVE:                      1429 (84.059%)
      -Class 1 - CARDIOGENIC SHOCK:           110 ( 6.471%)
      -Class 2 - PULMONARY EDEMA:              18 ( 1.059%)
      -Class 3 - MYOCARDIAL RUPTURE:           54 ( 3.176%)
      -Class 4 - CONGESTIVE HEART FAILURE:     23 ( 1.353%)
      -Class 5 - THROMBOEMBOLISM:              12 ( 0.706%)
      -Class 6 - ASYSTOLE:                     27 ( 1.588%)
      -Class 7 - VENTRICULAR FIBRILLATION:     27 ( 1.588%)
================================================================================
SAMPLES DISTRIBUTION BY: LET_IS (BINARY)
================================================================================
      -Class 0 - ALIVE:                      1429 (84.059%)
      -Class 1 - DEAD:                        271 (15.941%)
================================================================================
```

As reported, the dataset suffers of severe imbalance with respect to every single possible targets. In particular, the LET_IS target, even if we choose to collapse in just two class (ALIVE / DEAD), still remains with a severe imbalance.
Whatever is the target we choose to use to perform the analysis, we should address the imbalance with different techniques (data-level, algorithm-level, cost-sensitive-learning). We will address this problem in the modelling phase.

DATA UNDERSTANDING: *Data Exploration Report*

- **Class Balance:** From the class balance analysis of both all the possible complications and lethal outcome (binary / multi class), the dataset suffers of severe imbalance. In the modelling phase we will address the imbalance with different techniques (data-level, algorithm-level, cost-sensitive-learning).





- **Quantitative Analysis**
  - **Quantitative Discrete Variables**
    - **AGE:** variable tracks 1,692 individuals averaging **61.86 years** old, with a total range extending from **26 to 92**. Its slight negative skewness of **-0.22** and the close proximity between the mean and the **63.00 median** reflect a stable distribution where half of the group falls between **54 and 70** years. Histogram, KDE, and probability plots further validate this **near-normal, unimodal shape**, showing only minor deviations from a perfect bell curve at the extreme ends of the age spectrum.
    - **S_AD_KBRIG:** variable tracks 624 entries with an average value of **136.91** and a median of **140.00**, suggesting a highly balanced dataset. While the total range spans from **0 to 260**, most values are tightly packed within the **120 to 160** interquartile range. With a negligible skewness of **0.09**, the distribution presents as nearly symmetrical; this is visually confirmed by the clean, unimodal bell curve in the KDE and the consistent tracking of the theoretical normal line in the probability plot.

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- **D_AD_KBRIG:** variable includes 624 entries with a mean value of **81.39** and a median of **80.00**, indicating a central tendency that is closely aligned. The data spans a range from **0 to 190**, though the majority of observations are concentrated between **70 and 90**, as shown by the interquartile range. With a negative skewness of **-0.75**, the distribution leans slightly toward higher values, a characteristic confirmed by the KDE plot and the visible "steps" in the probability plot where the observed values deviate from the theoretical normal line at both the lower and higher extremes.
- **S_AD_ORIT:** variable contains **1,433** observations with a mean value of **134.59** and a median of **130.00**, indicating a fairly central distribution. The data ranges from a minimum of **0.0** to a maximum of **260.0**, with the middle 50% of values falling between **120.0 and 150.0**. A negative skewness of **-0.28** suggests a slight lean towards higher values, which is reflected in the KDE plot's nearly symmetrical bell shape. While the probability plot shows the data following the normal line through the center, visible deviations at the lower and upper tails indicate some variance at the extreme ends of the spectrum.
- **D_AD_ORIT:** variable consists of **1,433** entries with a mean of **82.75** and a matching median of **80.00**, indicating a strong central concentration. While the full range spans from **0 to 190**, most data points are clustered between **80 and 90**. With a negative skewness of **-1.03**, the distribution leans heavily toward higher values. This skew is clearly visible in the KDE plot’s sharp peak and the probability plot, where significant "stepping" away from the theoretical normal line at both ends confirms a non-normal distribution with distinct outliers at the lower extreme.

- **Quantitative Continuous Variables**
  - **K_BLOOD:** variable consists of **1,329** observations, exhibiting a mean of **4.19** and a median of **4.10**, which indicates a central concentration of data points. The values span from a minimum of **2.30** to a maximum of **8.20**, with the middle 50% of the population falling within the narrow interquartile range of **3.70 to 4.60**. With a positive skewness of **0.96**, the distribution is clearly right-skewed; this is visually confirmed by the elongated tail in the KDE plot and the probability plot, where observed values curve sharply away from the theoretical normal line at the higher end, indicating the presence of outliers.
  - **NA_BLOOD:** variable tracks **1,325** entries with a mean of **136.55** and a nearly identical median of **136.00**, indicating a well-balanced central concentration. While observations span from **117 to 169**, the majority of the data is tightly clustered between **133 and 140**. With a low positive skewness of **0.12**, the distribution is largely symmetrical; this is visually supported by the unimodal bell curve in the KDE plot and the way observed values closely follow the theoretical normal line in the probability plot, despite some minor deviations at the extreme ends.
  - **ALT_BLOOD:** variable comprises **1,416** observations, featuring a mean of **0.48** and a lower median of **0.38**, which suggests a concentration of data points at the lower end of the scale. The values extend from a minimum of **0.03** to a maximum of **3.00**, with the interquartile range showing that 50% of the sample falls between **0.23 and 0.61**. With a significant positive skewness of **2.27**, the distribution is heavily right-skewed; this is visually evident in the KDE plot's long right tail and the probability plot, where the data points curve sharply away from the theoretical normal line as values increase, indicating a substantial presence of high-value outliers.
  - **AST_BLOOD:** variable tracks **1,415** entries with a mean of **0.26** and a median of **0.22**, showing a concentration of data points at the lower end of the spectrum. While the total range extends from **0.04 to 2.15**, the interquartile range indicates that half of the observations are clustered between **0.15 and 0.33**. With a high positive skewness of **2.56**, the distribution is heavily right-skewed; this is visually confirmed by the KDE plot's sharp peak followed by a long right tail and the probability plot, where the data points curve steeply away from the theoretical normal line at the higher extreme, indicating significant outliers.
  - **KFK_BLOOD:** variable represents a very small sample size of only **4 observations**, with a mean value of **2.00** and a median of **1.35**. The values span from **1.20 to 3.60**, with a standard deviation of **1.10** indicating a fair amount of spread relative to the scale. Given the positive skewness of **1.70**, the distribution is right-skewed, which is visible in the KDE plot as a peak around the lower values followed by a tail extending toward the maximum. The probability

plot further highlights the non-normality caused by the small sample size, showing significant gaps and deviations between the observed data points and the theoretical normal line.

  - **L_BLOOD:** variable consists of **1,575** observations with a mean value of **8.78** and a median of **8.00**, indicating that the average is pulled slightly higher than the center of the data. The measurements range from a minimum of **2.00** to a maximum of **27.90**, with half of the entries clustered between **6.40 and 10.45**. With a positive skewness of **1.34**, the distribution is right-skewed; this is visually represented by the extended right tail in the KDE plot and the probability plot, where the data points curve away from the theoretical normal line at the higher end, highlighting the presence of several high-value outliers.
  - **ROE:** variable contains **1,497** observations with a mean of **13.44** and a median of **10.00**, indicating that a few high values are pulling the average upward. While the data range is broad, spanning from **1.00 to 140.00**, half of the sample is concentrated between **5.00 and 18.00**. With a significant positive skewness of **2.29**, the distribution is heavily right-skewed; this is visually confirmed by the sharp peak at lower values followed by a very long tail in the KDE plot and a probability plot that curves steeply away from the theoretical normal line due to extreme outliers at the upper end.

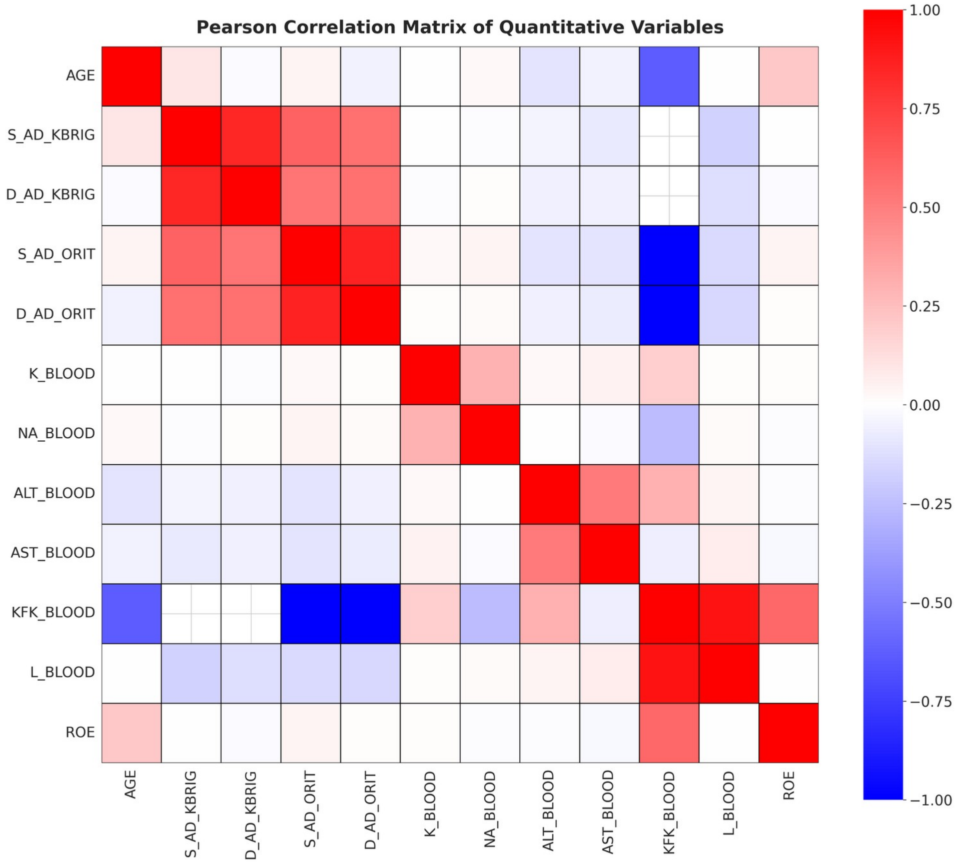
- **Consideration on Variables Distribution and Possible Transformation**

  - **Features that *do* deserve a transformation (e.g., `log1p`):** Use `log1p` when the variable is strictly non-negative and shows strong right skew / long upper tail (your hist + Q–Q curvature confirms this).
    - **Strong candidates (clear win):**
      - **ROE** (very heavy right tail, extreme outlier up to ~140)
      - **L_BLOOD** (right-skew with high outliers up to ~28)
      - **ALT_BLOOD** (strong right skew)
      - **AST_BLOOD** (strong right skew)
    - **Optional / borderline:**
      - **K_BLOOD** (moderate right skew; `log1p` can help a bit, but the distribution is already fairly compact. You can also leave it untransformed and just scale.)
    - **Do NOT bother:**
      - **KFK_BLOOD** (n=4 → too sparse; typically drop or treat as “mostly missing”, not transform)
  - **Features that *do not* need log-transform:** Don’t transform if the variable is already close to symmetric / near-Gaussian.
    - **NA_BLOOD** (skew ~0.12; Q–Q fairly linear) → **no log**
    - **AGE** → **no log** (already close to normal-ish)
    - **S_AD_KBRIG / S_AD_ORIT** → usually **no log** (roughly bell-shaped; check outliers only)
    - **D_AD_KBRIG / D_AD_ORIT** → **no log** *until you fix the zeros* (those zeros likely encode missing/invalid and create artificial skew)
  - **`StandardScaler`** on all continuous numeric features (including the log-transformed ones)

- **Correlation Matrix**
  - These 4 variables carry largely overlapping information → for linear models this is classic **multicollinearity**:
    - **S_AD_KBRIG ↔ D_AD_KBRIG is very high (deep red).**
    - **S_AD_ORIT ↔ D_AD_ORIT** is also **very high**.
    - Cross-correlations between **KBRIG** and **ORIT** BP variables are **moderate–high**.
  - Not as extreme as BP, but still partially redundant; usually fine to keep both, but we can also consider ratios (AST/ALT) depending on your clinical framing:
    - **ALT_BLOOD ↔ AST_BLOOD** shows a **clear positive correlation** (as expected).
  - Little linear redundancy here; scaling/transform choices matter more than collinearity:
    - **K_BLOOD** and **NA_BLOOD** look **weakly/moderately** related at most.
    - **AGE** has **low correlation** with most variables (weak linear relationships).

- **ROE** appears mostly weakly related to most variables.

- **KFK_BLOOD** is almost certainly misleading (only 4 record).



DATA PREPARATION: *Data Sampling Report*

In the Data Sampling Phase we define the different dataset for the different ICU time of analysis. For each time slot (admission, at the end of the first day, at the end of the second day and at the end of the third day), some features are removed from the analysis, as follow:

- **admission**
  - **Excluded:** R_AB_1_n, R_AB_2_n, R_AB_3_n, NA_R_1_n, NA_R_2_n, NA_R_3_n, NOT_NA_1_n, NOT_NA_2_n, NOT_NA_3_n
  - **Quantity:** Rows (1700) - Columns (115)

- **end of first day (after 24 hours)**
  - **Excluded:** R_AB_2_n, R_AB_3_n, NA_R_2_n, NA_R_3_n, NOT_NA_2_n, NOT_NA_3_n
  - **Quantity:** Rows (1700) - Columns (118)

- **end of second day (after 48 hours)**
  - **Excluded:** R_AB_3_n, NA_R_3_n, NOT_NA_3_n
  - **Quantity:** Rows (1700) - Columns (121)

- **end of third day (after 72 hours)**
  - **Excluded: -**
  - **Quantity:** Rows (1700) - Columns (124)

Additionally, we add for each time slot a new possible target column called "LET_IS_BINARY", for which the value 0=ALIVE remains the same, while all the possible class with value greater or equal than 1 are converted in 1=DEAD.

So in the end we have 4 dataset with the following shape:

- **admission**
  - **Excluded:** R_AB_1_n, R_AB_2_n, R_AB_3_n, NA_R_1_n, NA_R_2_n, NA_R_3_n, NOT_NA_1_n, NOT_NA_2_n, NOT_NA_3_n
  - **Quantity:** Rows (1700) - Columns (116)

- **end of first day (after 24 hours)**
  - **Excluded:** R_AB_2_n, R_AB_3_n, NA_R_2_n, NA_R_3_n, NOT_NA_2_n, NOT_NA_3_n
  - **Quantity:** Rows (1700) - Columns (119)

- **end of second day (after 48 hours)**
  - **Excluded:** R_AB_3_n, NA_R_3_n, NOT_NA_3_n
  - **Quantity:** Rows (1700) - Columns (122)

- **end of third day (after 72 hours)**
  - **Excluded: -**
  - **Quantity:** Rows (1700) - Columns (125)

For the modelling task we select the ICU time slot corresponding to the admission, so for the next parts we will focus just on using this dataset.

DATA PREPARATION: ***Data Cleaning Report***

The initial unclean dataset has the following shape (1700 rows, 116 columns ID included, 115 ID excluded). In the data quality inspection phase we discover that there were a lot of missing values in the dataset. Thus, in the cleaning phase we need to handle this issue at different level.

At row level, we know that on average, each record has ~ 7.6% of missing values. So, we fix a threshold of missingness at 20% and we chose to remove from the dataset all the record with more that 20% of missing values. At this point, 128 records were removed from the dataset and the remaining number of records is now equal to 1572.

At column level we act in two phase. First, we remove all the column with a quantity of missing values greater than 30%. In this way we passed from 116 columns ID included (115 ID excluded) to 109 columns ID included (108 ID excluded).
The 7 deleted columns were: ISB_NASL, S_AS_KBRIG, D_AD_KBRIG, KFK_BLOOD, NA_KB, NOT_NA_KB, LID_KB.

Secondly, we select just one target variable (LET_IS_BINARY) to focus on, and we drop all the other 12 possible targets: FIBR_PREDS, PREDS_TAH, JELUD_TAH, FIBR_JELUD, A_V_BLOK, OTEK_LANC, RAZRIV, DRESSLER, ZSN, REC_IM, P_IM_STEN, LET_IS.

At the end of the cleaning phase the final dataset has the following shape (1572 rows, 97 columns ID included, 96 ID excluded).
Remaining columns: ID, AGE, SEX, INF_ANAM, STENOK_AN, FK_STENOK, IBS_POST, GB, SIM_GIPERT, DLIT_AG, ZSN_A, nr_11, nr_01, nr_02, nr_03, nr_04, nr_07, nr_08, np_01, np_04, np_05, np_07, np_08, np_09, np_10, endocr_01, endocr_02, endocr_03, zab_leg_01, zab_leg_02, zab_leg_03, zab_leg_04, zab_leg_06, S_AD_ORIT, D_AD_ORIT, O_L_POST, K_SH_POST, MP_TP_POST, SVT_POST, GT_POST, FIB_G_POST, ant_im, lat_im, inf_im, post_im, IM_PG_P, ritm_ecg_p_01, ritm_ecg_p_02, ritm_ecg_p_04, ritm_ecg_p_06, ritm_ecg_p_07, ritm_ecg_p_08, n_r_ecg_p_01, n_r_ecg_p_02, n_r_ecg_p_03, n_r_ecg_p_04, n_r_ecg_p_05, n_r_ecg_p_06, n_r_ecg_p_08, n_r_ecg_p_09, n_r_ecg_p_10, n_p_ecg_p_01, n_p_ecg_p_03, n_p_ecg_p_04, n_p_ecg_p_05, n_p_ecg_p_06, n_p_ecg_p_07, n_p_ecg_p_08, n_p_ecg_p_09, n_p_ecg_p_10, n_p_ecg_p_11, n_p_ecg_p_12, fibr_ter_01, fibr_ter_02, fibr_ter_03, fibr_ter_05, fibr_ter_06, fibr_ter_07, fibr_ter_08, GIPO_K, K_BLOOD, GIPER_NA, NA_BLOOD, ALT_BLOOD, AST_BLOOD, L_BLOOD, ROE, TIME_B_S, NITR_S, LID_S_n, B_BLOK_S_n, ANT_CA_S_n, GEPAR_S_n, ASP_S_n, TIKL_S_n, TRENT_S_n, LET_IS_BINARY

All the remaining missing values in the dataset will be handled by imputation during the modelling phase / training pipeline to avoid data leakage and bias injection.

DATA PREPARATION: ***Feature Selection Report***

DATA PREPARATION: ***Data Construction Report***

DATA PREPARATION: ***Data Integration Report***
NOTHING TO REPORT

DATA PREPARATION: ***Data Format Report***
NOTHING TO REPORT

MODELING
NOTHING TO REPORT

EVALUATION
NOTHING TO REPORT

DEPLOYMENT
NOTHING TO REPORT